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Observations on the Behavior of a Liquid Droplet Transported by a Gas Stream Impinging on an Isothermal Surface

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Abstract

Droplets impinging on an isothermal surface have been extensively investigated both computationally and experimentally. The current work is related to the behavior of a droplet that is transported to the target surface by a gas stream that impinges on an isothermal surface. The gas stream not only transports the droplet to the surface, it also contributes to the droplet spreading dynamics. This paper will report on preliminary experiments and numerical simulations of this unique problem, which has heretofore not been reported in the literature. Experiments and computational simulations using a volume of fluid (VOF) method have been performed for a single water droplet in a circular air jet impinging on an isothermal (non-heated) surface. The primary scaling parameters affecting the problem are the droplet and jet Reynolds number (Re_j, Re_D), and the nozzle to plate spacing (H/D_j). High-speed photography was used to capture the droplet spreading dynamics and the results compared with computational simulations with the carrier jet velocity in the laminar regime. It is shown that the gas jet profoundly affects the spreading dynamics by imposing pressure and strain on the droplet surfaces, causing the maximum spreading diameter to increase in a linear fashion.

Keywords: Droplet, Impinging Jet, Isothermal.

Nomenclature

D_0	Droplet diameter, mm
D_j	Nozzle diameter, mm
D_{max}	Maximum spreading diameter, mm
D_s	Spreading diameter, mm
H	Nozzle to plate spacing, mm
d_n	Hypodermic needle diameter, mm
d_p	Aluminum plate diameter, mm
d_t	Tube diameter, mm
t_s	Maximum spreading time, ms
v_0	Droplet velocity, m/s
v_{jet}	Carrier jet velocity, m/s

Greek Symbols

θ	Contact angle
μ	Kinematic viscosity, kg/m·s
ρ	Density, kg/m^3
σ	Surface tension, N·m

Non-dimensional Numbers

Re_D	Reynolds number of the drop, $[\rho_{air} v_0 D_0/\mu_{air}]$
Re_j	Reynolds number of the jet, $[\rho_{air} v_{jet} D_j/\mu_{air}]$
We	Weber number, $[\rho v_0 D_0/\sigma]$
τ	Spreading time, $[t_s v_0/D_0]$

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1. Introduction

The physics of a droplet impinging on a surface in a stagnant air environment are important to industrial processes from rapid prototyping to spray cooling of electronics. The investigation into the dynamics of a droplet impinging on a flat, isothermal surface has been extensively investigated. The review by Yarin [1] summarizes the previous work and as such only the most relevant studies are briefly discussed here. There have been no previous investigations of a droplet that is transported to a surface by a gas jet impinging on the surface. This is a unique problem that has been previously unreported.

The experimental characterization of the problem of a single droplet impinging on a surface was carried out first by Chandra and Avedisian [2] using high-speed photography to capture the spreading process of the drop impinging on a substrate. This work was furthered by Roisman et al. [3], in which an analytical formulation of the problem was also developed. Roisman et al. found that the structure of the droplet upon initial impact resembled a hemispherical core structure surrounded by a ring structure, which advanced as time increased. The region between the collapsing core and the advancing ring is termed the droplet lamella and its behavior is dependent on the surface roughness, characterized by the contact angle. The experimental work on the subject showed the behavior of a droplet impinging on a surface to be governed by the impact velocity, the initial drop size, and the surface tension effects [4].

Hatta et al. [5] used a MAC-type solution method to model the deformation of a liquid water droplet on a solid surface, and while the computational results mirrored those of the experimental results, the surface tension effects were neglected. The first complete numerical characterization of the problem was performed by Fukai et al. [6]; all previous works had neglected the viscous forces in the problem, but their Lagrangian finite-element model captured the physics completely with the exception of the relationship between the contact angle and the lateral velocity of the contact line. Neglecting the capillary force at the contact line was found to be valid for high impact velocities. In the case where the droplet struck the surface with a velocity due only to the gravitational acceleration of its own mass, the radial flow reversed direction showing the effects of the surface tension in pulling the droplet toward its equilibrium state. A semi-empirical model based on the results of Fukai et al. [7] to determine the maximum spreading diameter was found by correlating the numerical results with the dimensionless groups of the dynamic and contact angle. Bussman et al later modeled the splashing effects that were often neglected in the numerical models using an enhanced RIPPLE model [8]. The model accurately depicted the fingering and splashing of various liquid droplets for a wide range of parametric conditions. The use of the VOF model to characterize the surface of the deforming liquid droplet rather than the finite element method has been employed since the introduction of the method in a widespread fashion [9-13]. Kamnis et al. [14] used a VOF model that yielded results that were very accurate and computationally less demanding than the FEA methods of previous investigations.

The most recent work on the subject has been concerned with comparing the use of the VOF model with experimental results found using high-speed photography. Mehdi-Nejad et al. [15] compared experimental and computational results for various fluids impacting on different materials, finding an air bubble at the solid liquid interface. This was due to an increase in the pressure at the interface that depressed the droplet surface. Gunjal et al. [16] investigated the dynamics of a water droplet spreading and recoiling on glass and Teflon at different Reynolds numbers and Weber numbers. The work found the contact angle, θ, to vary with time during the spreading process; microscopic factors such as surface roughness, surface absorption, and variation in surface tension were found to greatly affect the spreading of the droplet.

In this work, a water droplet is transported towards the test surface by an air jet. The jet facilitates the spreading of the liquid film following droplet impact. Because water is the only fluid used in this work, the Weber number, We, is constant. In addition the jet to target spacing, H/D_j, was held constant.

2. Experimental and Computational Approach

2.1. Experimental Apparatus

The experiment, shown in Fig. 1, was designed to establish an idealized situation of a droplet transported towards the target surface by a uniform, stagnation flow of a gas. To create the flow, an axisymmetric, converging nozzle with a 36:1 area ratio was utilized as shown. The nozzle exit diameter, D_j, was 25.4mm and measurement of the air velocity at the exit plane by traversing a total pressure tube showed that the local jet velocity was uniform to within 0.02% of the mean exit velocity.

A simple droplet generator was fabricated from a 6.35 mm diameter stainless steel tube, d_t, ending with a 0.94 mm hypodermic needle, d_n. Its tip is placed 10 cm above the jet exit and it is connected to a simple syringe that is used to pump a droplet out of the needle. The impinging jet nozzle creates a uniform flow velocity profile, and its exit plane is positioned so that H/D_j is 2. This ensures that the 25.4 mm aluminum surface, d_p, is within the limits defining a confined impinging jet.

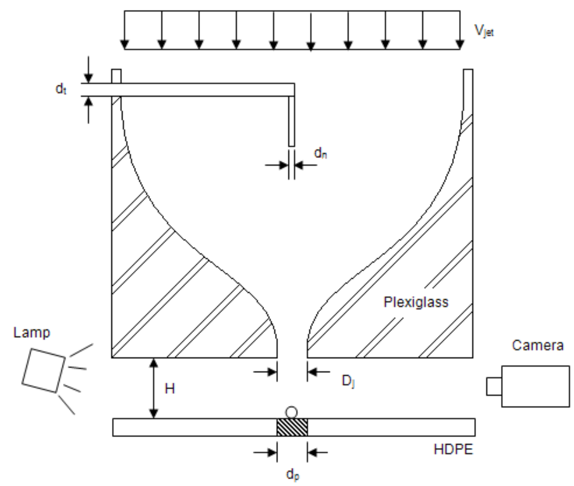

Fig. 1. Experimental Apparatus Schematic

One possible experimental problem that may be present is the probable existence of a hydrodynamic wake downstream of the injector. It is quite likely that the droplet is injected into this low momentum injector wake, thereby preventing the droplet from achieving the same velocity as the main gas jet.

A Vision Research Phantom Miro 2, monochrome camera was used to capture the droplet traveling to, impinging upon, and spreading on the surface. The photography was performed by backlighting the droplet with a halogen lamp equipped with a light diffuser. The lamp was aligned with the camera lens, and images were recorded using the camera software.

Table 1. Data collection points and camera frame rate

Re_j (-)	Framing Rate (fps)
0	10000
5029	10000
8365	13000
11860	15000
14860	15000
18311	15000
21290	15000
25263	19000
28348	19000

The various experimental conditions investigated are shown in Table 1, along with the different framing rates used at each Re_j. The initial droplet diameter, D_0, was found with the imaging software by reference to the known actual diameter of the aluminum surface, 25.4 mm, and assuming that the droplet was initially spherical.

2.2. Computational simulation

The computational simulations were carried out using the standard k-ω turbulence model for a two-dimensional axisymmetric confined jet, and the modeling equations are given by Eqs. 1 and 2.

$$\frac{\partial}{\partial t}(\rho k)+\nabla\cdot(\rho k\vec{u})=\nabla\cdot(\Gamma_k\nabla k)+G_k-Y_k+S_k \qquad (1)$$

$$\frac{\partial}{\partial t}(\rho\omega)+\nabla\cdot(\rho\omega\vec{u})=\nabla\cdot(\Gamma_\omega\nabla\omega)+G_\omega-Y_\omega+S_\omega \qquad (2)$$

In Eqs. 1 and 2, ρ is the density, $\vec{u}$ is the velocity, k is the turbulence kinetic energy, ω is the specific dissipation rate, Γ_k and Γ_ω are the effective diffusivity of k and ω respectively, G_k is the generation of k due to mean velocity gradients, G_ω is the generation of ω, Y_k and Y_ω are the dissipation of k and ω due to turbulence, and S_k and S_ω are the source terms.

The interface between the liquid and gas is tracked using the solution to Eq. 3 where α_n is the n^{th} fluid's volume fraction.

$$\frac{\partial\alpha_n}{\partial t}+\vec{u}\cdot\nabla\alpha_n=0 \qquad (3)$$

The primary-phase volume fraction is computed based on Eq. 4 rather than Eq. 3.

$$\sum_{n=1}^{m}\alpha_n=1 \qquad (4)$$

The surface tension is included in the flow equations as a source term and the wall adhesion is incorporated by θ_w which is related to the surface normal as shown in Eq. 5 where $\hat{n}_w$ and $\hat{t}_w$ are the unit vectors normal and tangential to the wall respectively.

$$\hat{n}=\hat{n}_w\cos\theta_w+\hat{t}_w\sin\theta_w \qquad (5)$$

The initial condition (t=0) is set to be the moment when the water droplet first touches the aluminum surface. Table 2 shows the initial conditions for the six computational studies.

Table 2. Computational initial conditions

Case	1	2	3	4	5	6
Water						
v_0(m/s)	1.5	1.6	1.7	1.8	2.35	3.3
D_0(mm)	3.3	3.4	3.2	3.2	3.2	3.2
Air						
v_{jet}(m/s)	3	5	7.2	9	13.4	15
D_j(mm)	25.4	25.4	25.4	25.4	25.4	25.4

As with the experimental data reduction procedure, the droplets are assumed to be spherical. A uniform grid is used in the droplet vicinity and further than that a non-structured gird is implemented to diminish the computational requirements. Figure 2 shows a schematic of the boundary conditions employed in the impinging jet flow.

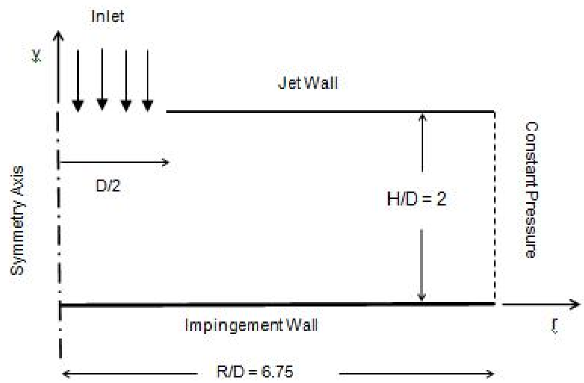

Fig. 2. Computational Domain

A total of six computational simulations were performed for varying Re_j. The fluid properties are summarized in Table 3.

Table 3. Computational fluid properties

Property	Water	Air
ρ (kg/m^3)	998.2	1.225
μ (kg/ms)	1.003E-3	1.791E-5
σ (N/m)	0.073	N/A
θ (°)	110	N/A

3. Results

The spreading time is normalized using Eq. 6, where the reference diameter and the reference velocity are taken prior to the droplet impinging on the surface.

$$\tau_{max}=\frac{t_s v_0}{D_0} \qquad (6)$$

The droplet velocity was determined from analysis of the digital video. As shown in Table 4, the droplet velocity was considerably smaller than the jet velocity. We hypothesize that this is due to droplet inertia as well as the fact that the drop is injected into the flow wake downstream of the injector. The contact angle for each case was evaluated and found to be constant as shown in Table 4.

Table 4. Experimental Results

Re_j	D_0 (mm)	v_0 (m/s)	Re_D	θ (°)
0	3.282	1.497	317	110.0
5029	3.256	1.527	321	110.0
8365	3.376	1.619	353	110.0
11860	3.215	1.709	354	110.0
14860	3.215	1.826	379	110.0
18311	2.170	2.196	307	110.0
21290	3.215	2.348	487	110.0
25263	3.215	2.669	553	110.0
28348	3.376	3.256	709	110.0

Figure 3 shows the computational results compared side by side with the experimental results. These results show the validity of the computational solution and the accuracy in the results. The spreading process can be seen in Fig. 4 where the droplet spreading diameter, D_s, is depicted as a function of τ for Re_j of 8365 and Re_D of 353.

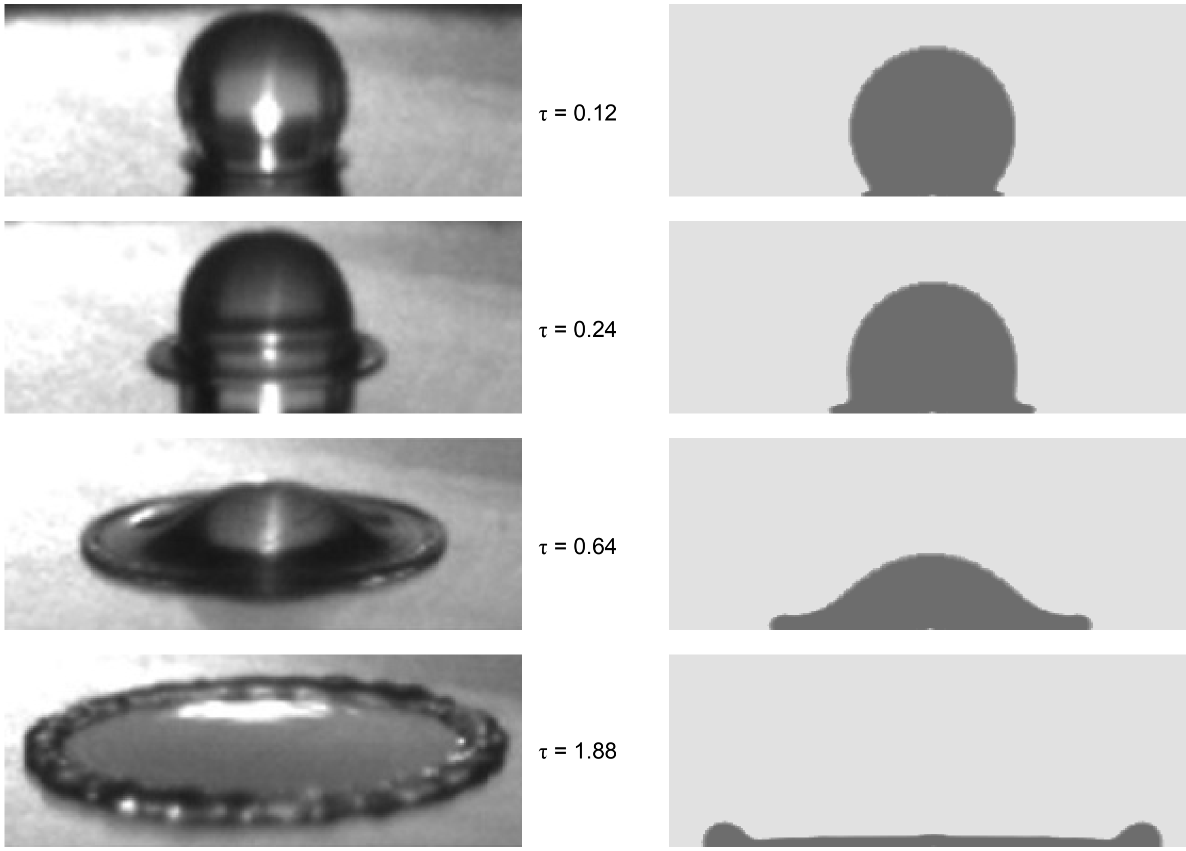

Fig. 3. Experimental & computational result comparison for (Re_j = 8365, Re_D = 353)

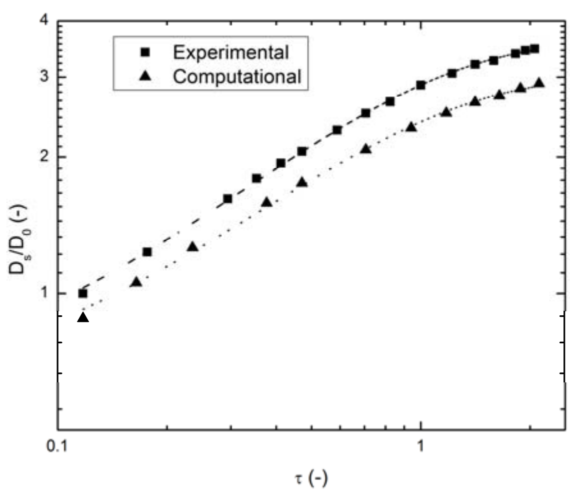

Fig. 4. Spreading Diameter Evolution (Re_j = 8365, Re_D = 353)

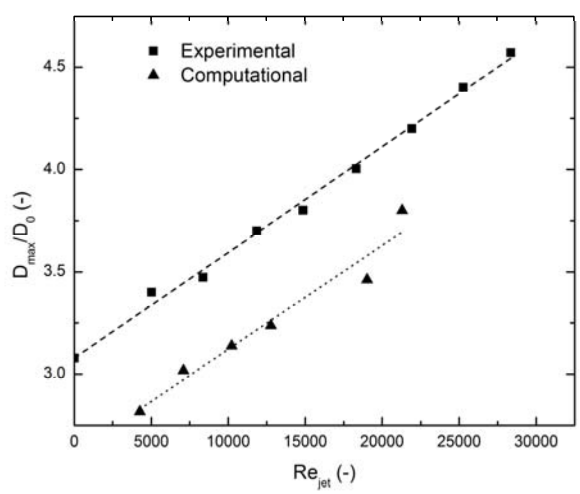

Fig. 5. Normalized maximum spread diameter

Figure 5 shows the effect of Re_j on the maximum spreading diameter of the droplet, D_{max}. The jet enhancement to the spreading problem is evident in the linearly increasing droplet spreading as Re_j increases. The normalized time is used to compare the maximum spreading times, i.e. the time to achieve maximum spreading at different carrier jet velocities, Fig. 6.

Figure 6 shows that the spreading time increases exponentially. It is believed that this behavior is due to the inertial effects of the impinging droplet dominating the spreading behavior, compared to the interfacial effects. Thus for a given droplet diameter, as the velocity of the droplet increases due to the increase in v_{jet}, t_s increases and this causes τ to increase.

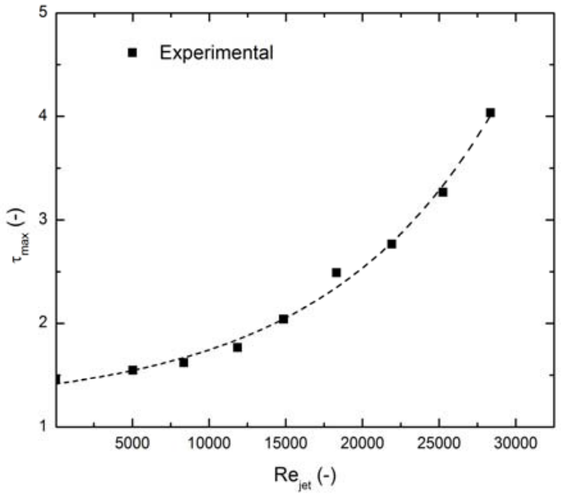

Fig. 6. Normalized maximum spreading time

4. Conclusions

The effects of a carrier gas on the droplet impingement dynamics were investigated both experimentally and computationally. High-speed photography was used to capture the droplet dynamics throughout the impingement process. Comparison of the experimental results with computational results found using a VOF model yielded good consistency between the two.

The effect of the carrier jet on the spreading of a liquid water droplet on an isothermal surface was to increase the droplet velocity. This led to an exponential increase in the normalized spreading time as well as a linear increase in the maximum spreading diameter of the droplet.

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